Seasonal Plots

Raphael Bihag

2026-03-26

Part 1: Loading Libraries

Create a chunk named setup, load necessary libraries

```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.4.3

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(urca)
```

```
## Warning: package 'urca' was built under R version 4.4.3
```

```
library(TSA)
```

```
## Warning: package 'TSA' was built under R version 4.4.3

## Registered S3 methods overwritten by 'TSA':
##   method       from
##   fitted.Arima forecast
##   plot.Arima   forecast

##
## Attaching package: 'TSA'

## The following objects are masked from 'package:stats':
##
##     acf, arima

## The following object is masked from 'package:utils':
##
##     tar
```

```
library(fpp3)
```

```
## Warning: package 'fpp3' was built under R version 4.4.3

## -- Attaching packages -------------------------------------------- fpp3 1.0.3 --
```

```
## v tibble      3.2.1     v tsibbledata 0.4.1
## v dplyr       1.2.0     v ggtime      0.2.0
## v tidyr       1.3.1     v feasts      0.5.0
## v lubridate   1.9.3     v fable       0.5.0
## v tsibble     1.2.0

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.2

## Warning: package 'lubridate' was built under R version 4.4.2

## Warning: package 'tsibble' was built under R version 4.4.3

## Warning: package 'tsibbledata' was built under R version 4.4.3

## Warning: package 'ggtime' was built under R version 4.4.3

## Warning: package 'feasts' was built under R version 4.4.3

## Warning: package 'fabletools' was built under R version 4.4.3

## Warning: package 'fable' was built under R version 4.4.3

## -- Conflicts ------------------------------------------------- fpp3_conflicts --
## x lubridate::date()    masks base::date()
## x dplyr::filter()      masks stats::filter()
## x tsibble::intersect() masks base::intersect()
## x tsibble::interval()  masks lubridate::interval()
## x dplyr::lag()         masks stats::lag()
## x tsibble::setdiff()   masks base::setdiff()
## x tsibble::union()     masks base::union()
```

```r
library(dplyr)
```

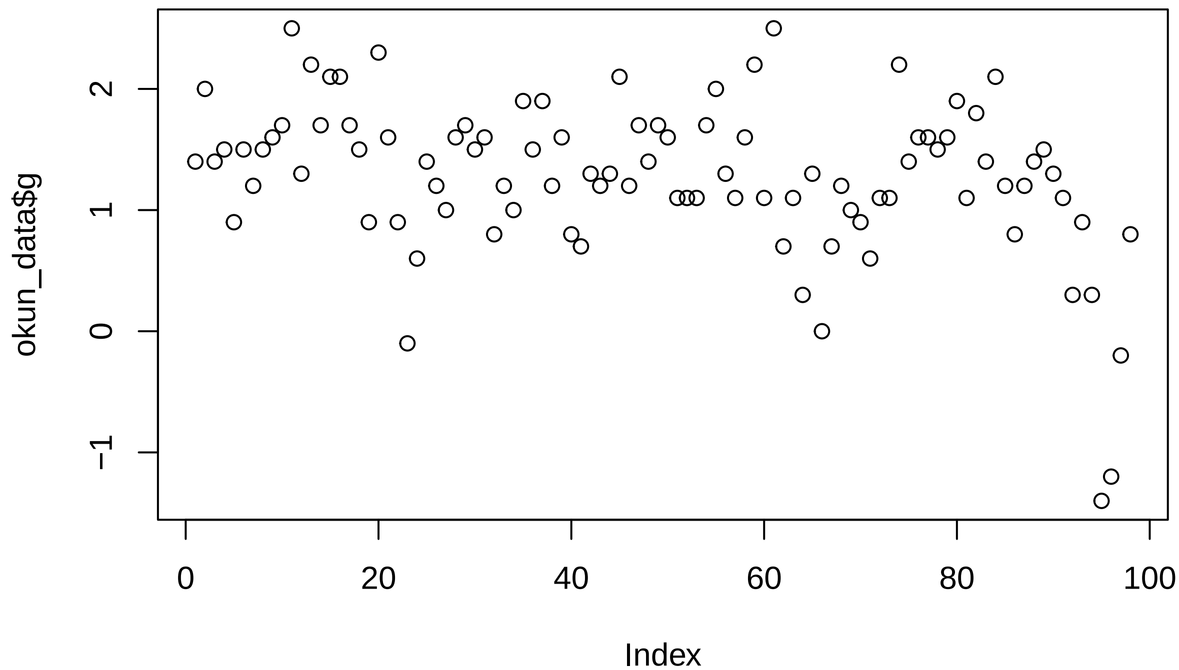
Import Dataset - GDP and unemployment rate

Data Info: A quarterly data (1985Q2 - 2009Q3) with 98 observations on the following 2 variables

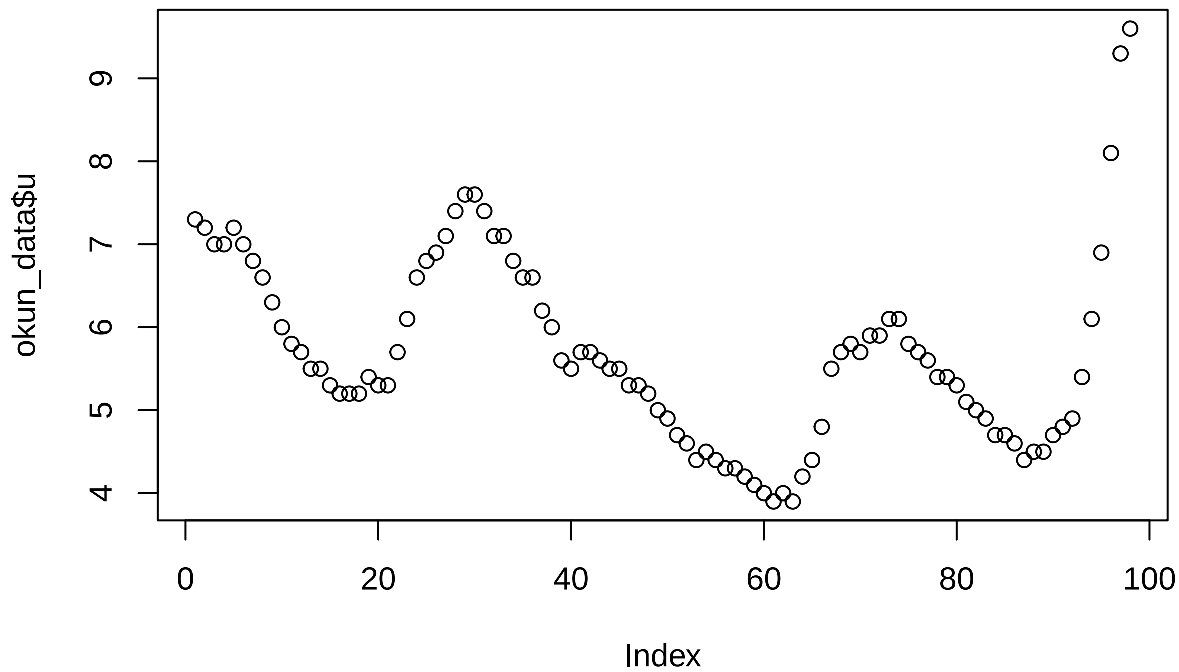
g – percentage change in U.S. Gross Domestic Product (seasonally adjusted)

u – U.S. Civilian Unemployment Rate (seasonally adjusted)

```r
# Import dataset
okun_data <- read.csv('C:/Users/Mary Grace Bihag/Desktop/Bayot Files/BSDS 1/2BSDS/AMAT 132/okun_data.cs

# Plot of data
plot(okun_data$g)
```




```r
plot(okun_data$u)
```

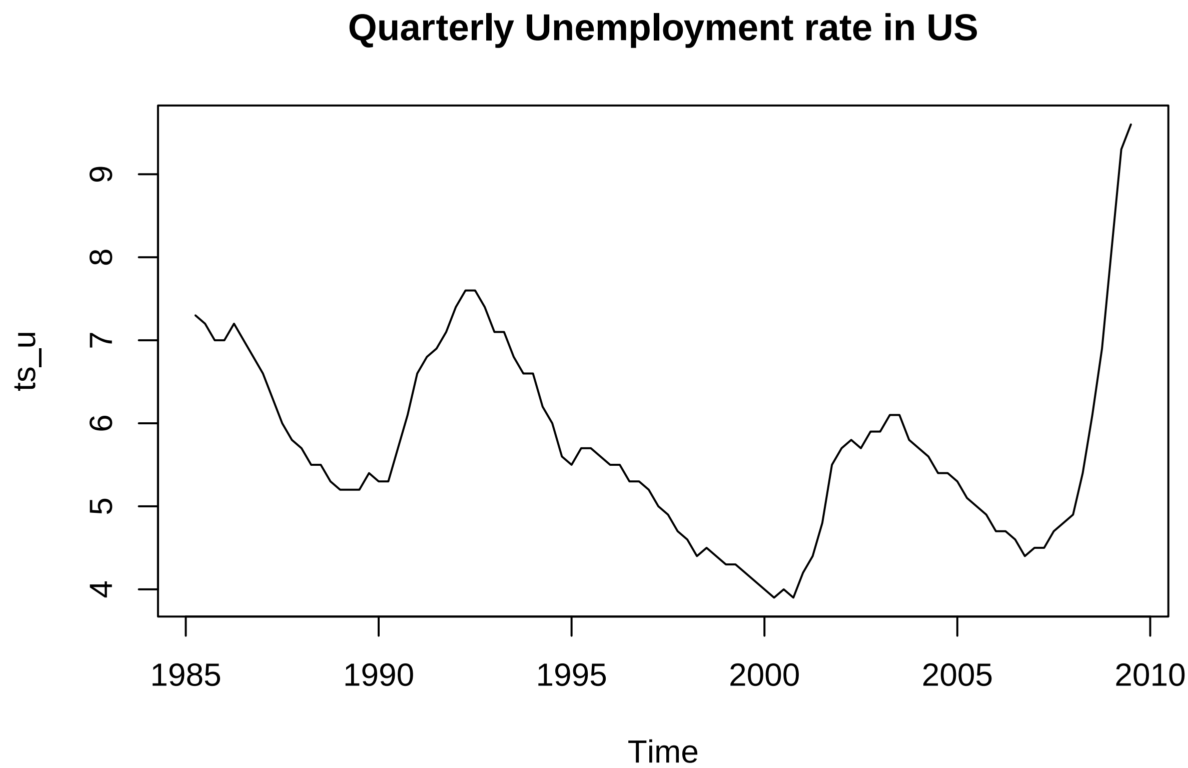



```r
# Step 1. Convert data to time series

help("ts")
```

```
## starting httpd help server ... done
```

```r
ts_g <- ts(data=okun_data$g, start=c(1985, 2), frequency=4)
ts_u <- ts(data=okun_data$u, start=c(1985, 2), frequency=4)

# Step 2. Time Series Plot
plot(ts_g, main="Quarterly GDP growth rate in US")
```


```r
plot(ts_u, main="Quarterly Unemployment rate in US")
```

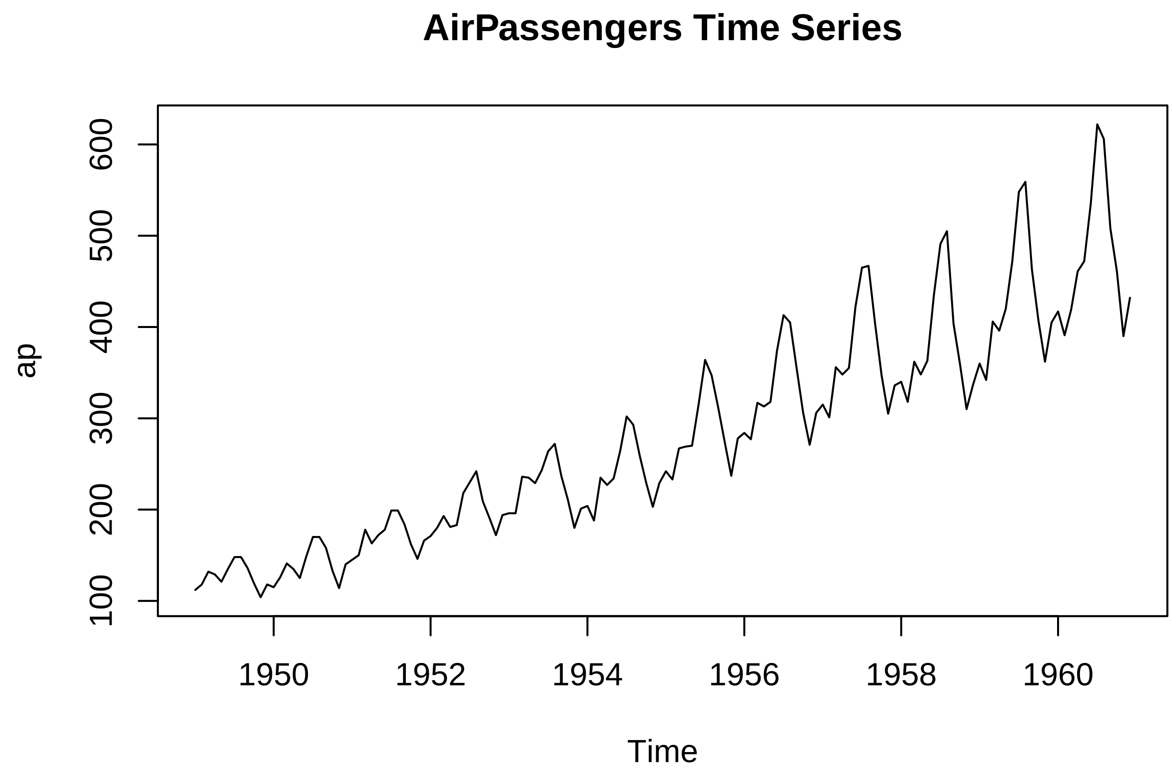


Example 1: AirPassengers Dataset

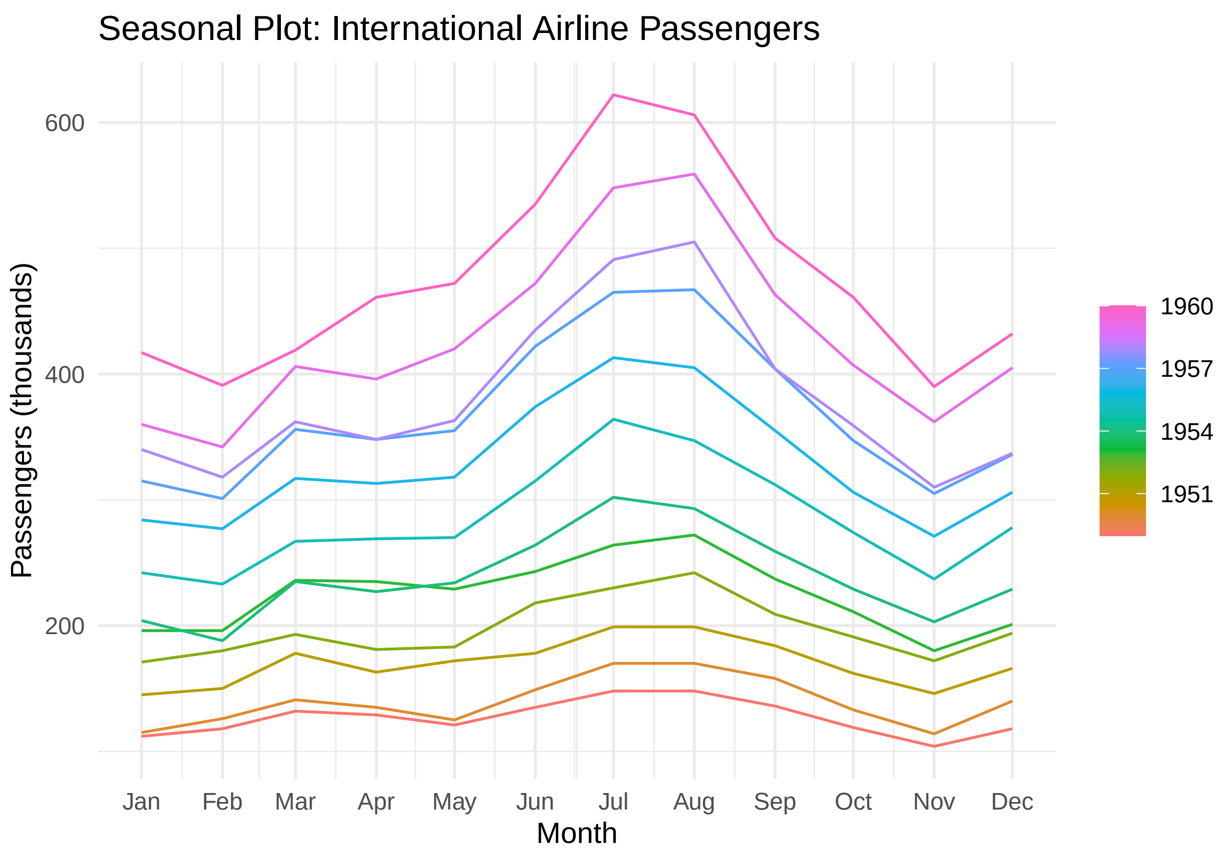
Data Info: Monthly Airline Passenger Numbers 1949-1960.

The classic Box & Jenkins airline data. Monthly totals of international airline passengers, 1949 to 1960.

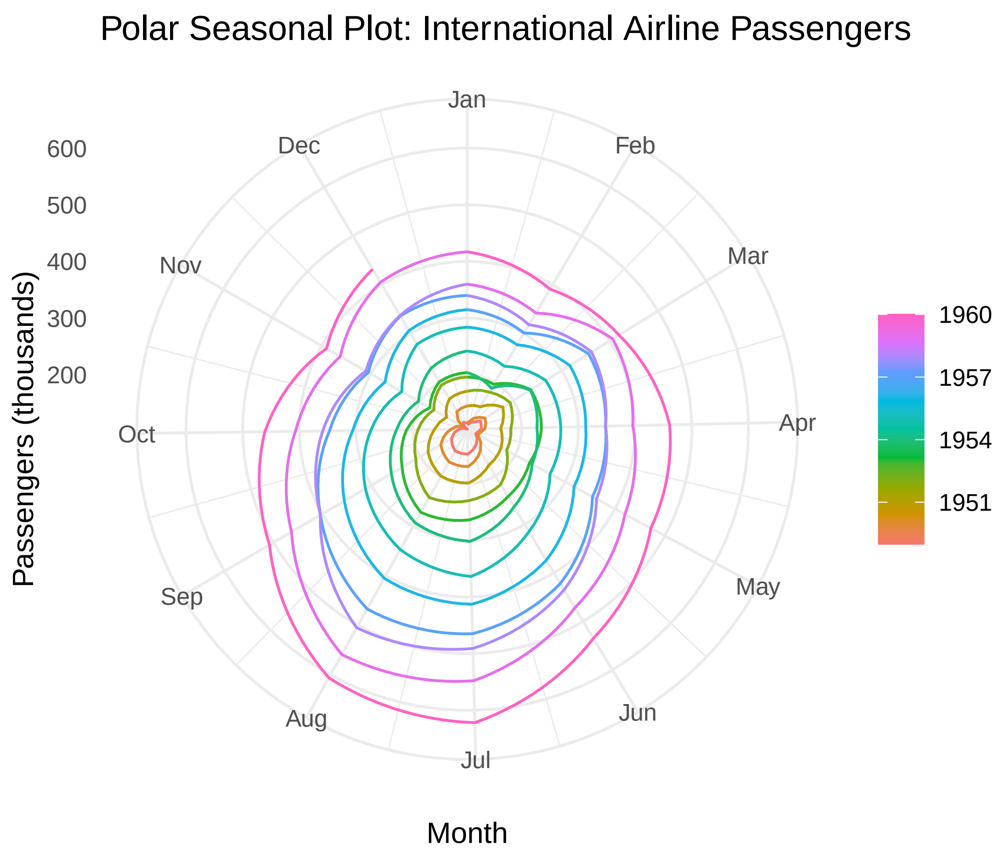
```r
# Step 1. Load dataset and time series plot
data("AirPassengers")
ap <- AirPassengers
plot(ap, main="AirPassengers Time Series")
```



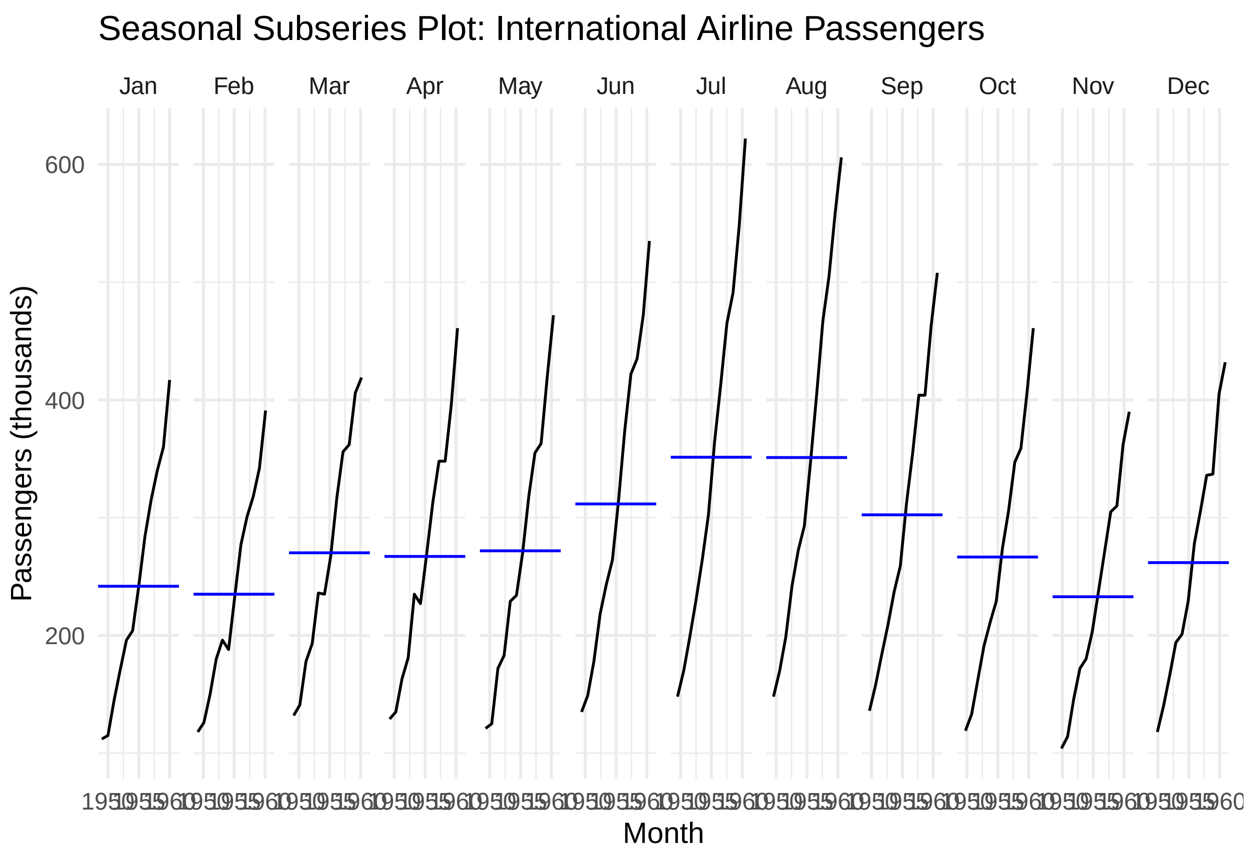

```r
# Step 2. Convert data to a tsibble object
# The tsibble format is preferred for the fpp3 functions
ap_tsibble <- ap %>%
as_tsibble()

# %>% is the pipe operator, takes the output of one function or operation and passes it as the first ar

# Step 3. Seasonal plot
ap_tsibble %>%
gg_season(value, period = "year") +
labs(
  title = "Seasonal Plot: International Airline Passengers",
  y = "Passengers (thousands)",
  x = "Month"
) +
theme_minimal()
```




```
# Step 4. Polar seasonal plot
ap_tsibble %>%
gg_season(value, period = "year", polar=TRUE) +
labs(
  title = "Polar Seasonal Plot: International Airline Passengers",
  y = "Passengers (thousands)",
  x = "Month"
) +
theme_minimal()
```




```r
# Step 5. Seasonal subseries plot
ap_tsibble %>%
gg_subseries(value) +
labs(
  title = "Seasonal Subseries Plot: International Airline Passengers",
  y = "Passengers (thousands)",
  x = "Month"
) +
theme_minimal()
```



Numerical Summaries

```r
# Mean
mean(ts_g)
```

```
## [1] 1.276531
```

```r
mean(ts_u)
```

```
## [1] 5.704082
```

```r
# Median
median(ts_g)
```

```
## [1] 1.3
```

```r
median(ts_u)
```

```
## [1] 5.5
```

```r
# Mean Absolute Deviations (MAD)
mean(abs(ts_g - mean(ts_g)))
```

```
## [1] 0.4564765
```

```r
mean(abs(ts_u - mean(ts_u)))
```

```
## [1] 0.8764265
```

```r
# Variance
var(ts_g)
```

```
## [1] 0.4185157
```

```r
var(ts_u)
```

```
## [1] 1.28287
```

```r
# Standard Deviation
sd(ts_g)
```

```
## [1] 0.6469279
```

```r
sd(ts_u)
```

```
## [1] 1.132638
```

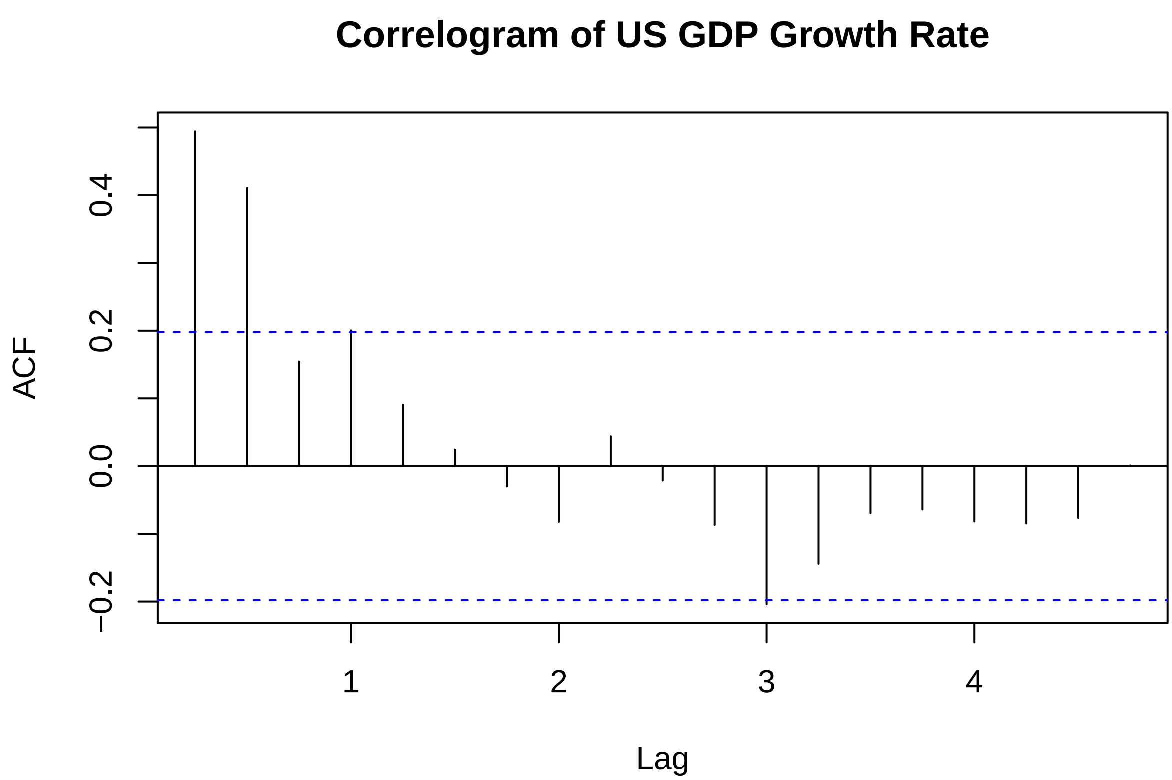
```r
# Covariance
cov(ts_g, ts_u)
```

```
## [1] -0.1775321
```

```r
# Correlation
cor(ts_g, ts_u)
```

```
## [1] -0.2422868
```

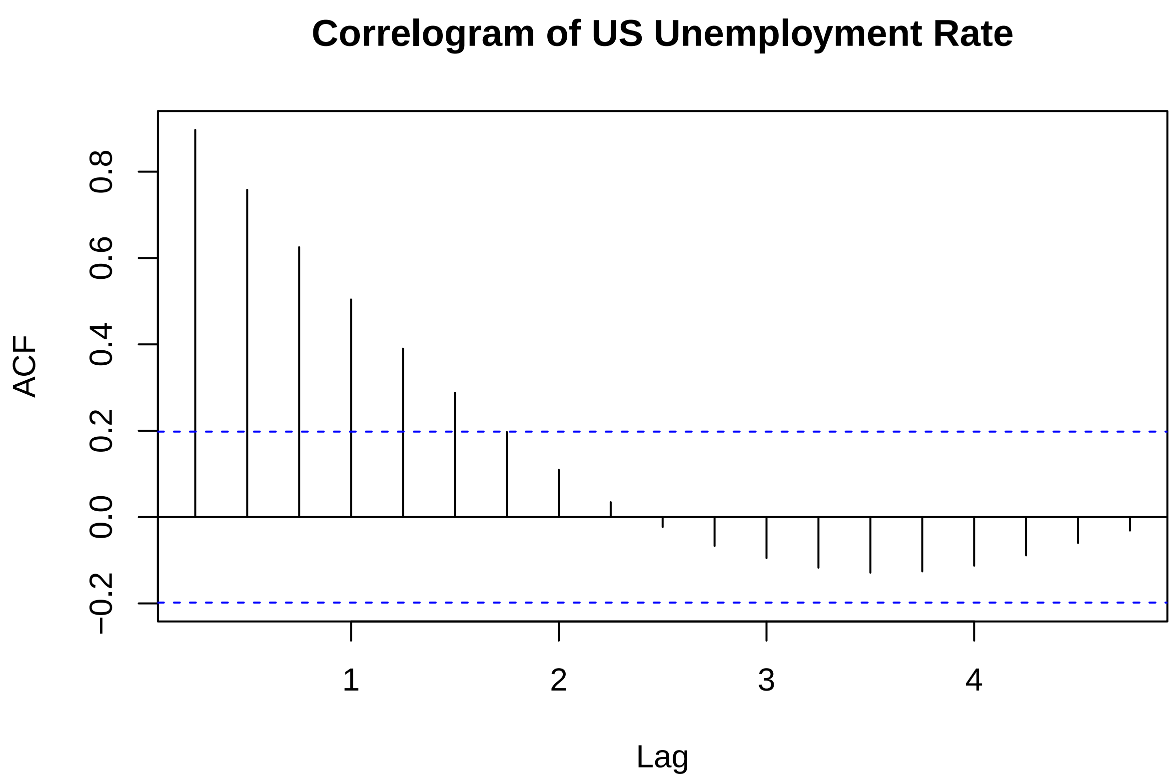
```r
# Autocorrelation and Correlogram
acf_values_g <- acf(ts_g, main = "Correlogram of US GDP Growth Rate")
```



```r
print(acf_values_g)
```

```
##
## Autocorrelations of series 'ts_g', by lag
##
##   0.25   0.50   0.75   1.00   1.25   1.50   1.75   2.00   2.25   2.50   2.75
##  0.494  0.411  0.154  0.200  0.090  0.024 -0.030 -0.082  0.044 -0.021 -0.087
##   3.00   3.25   3.50   3.75   4.00   4.25   4.50   4.75
## -0.204 -0.144 -0.069 -0.064 -0.082 -0.085 -0.077  0.001
```

```r
acf_values_u <- acf(ts_u, main = "Correlogram of US Unemployment Rate")
```




```r
print(acf_values_u)
```

```
##
## Autocorrelations of series 'ts_u', by lag
##
##   0.25   0.50   0.75   1.00   1.25   1.50   1.75   2.00   2.25   2.50   2.75
##  0.897  0.758  0.625  0.504  0.390  0.288  0.197  0.110  0.035 -0.023 -0.067
##   3.00   3.25   3.50   3.75   4.00   4.25   4.50   4.75
## -0.095 -0.117 -0.129 -0.126 -0.113 -0.089 -0.060 -0.031
```

The blue broken lines marks the significance area. In the example of the US GDP Growth Rate, we can interpret it as the previous two quarters have significant influence to the current quarter. Mathematically, $u_t = k_1 u_{t-1} + k_2 u_{t-2}$.

As for the US Unemployment Rate, its the same but we can say that the 6 previous quarters influence the current.